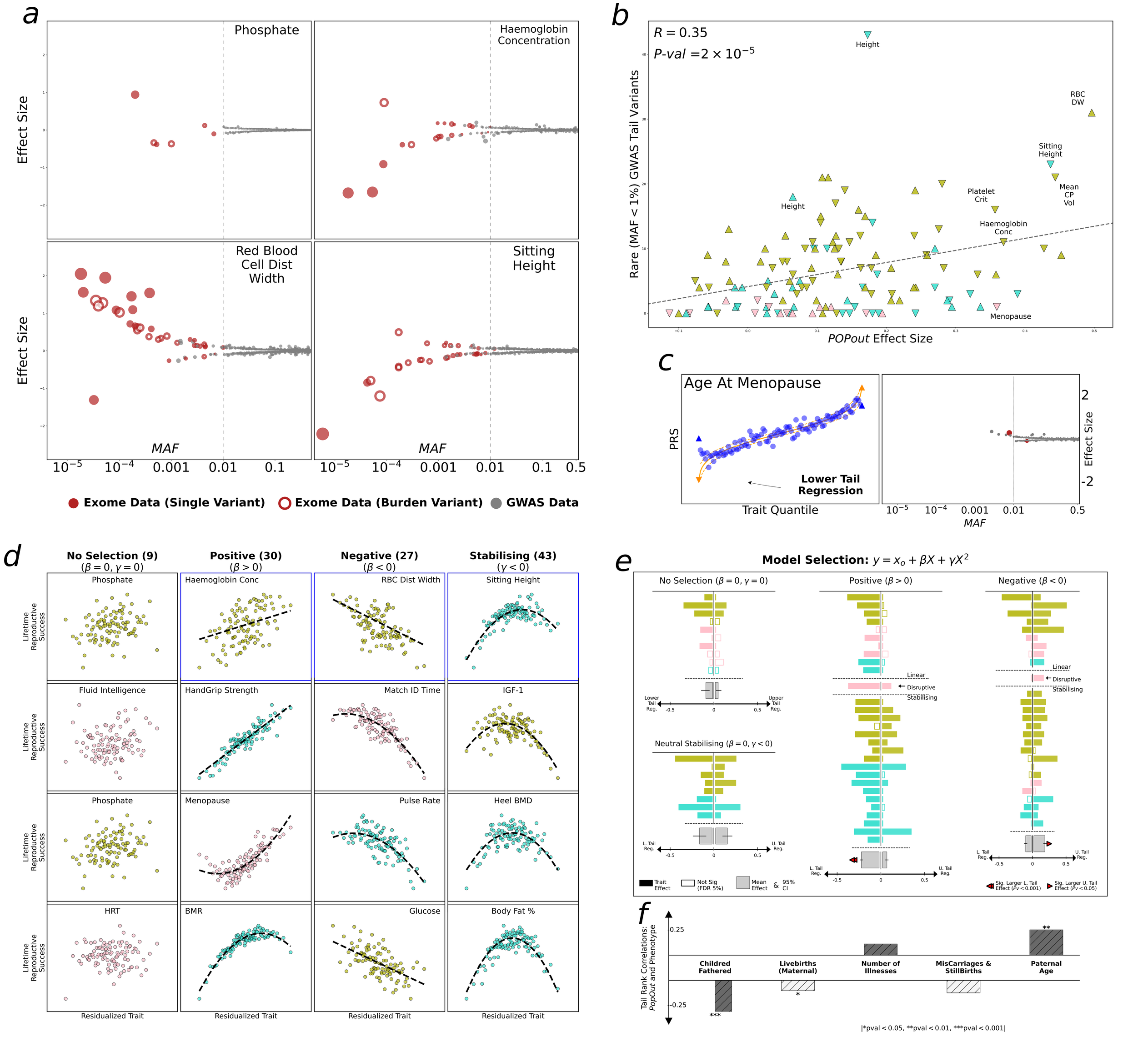
a
Phosphate
Haemoglobin Concentration
Red Blood Cell Dist Width
Sitting Height
Effect Size
MAF
Exome Data (Single Variant)
Exome Data (Burden Variant)
GWAS Data
b
R = 0.35
P-val = 2 × 10^−5
Height
RBC DW
Sitting Height
Mean CP Vol
Platelet Crit
Haemoglobin Conc
Menopause
Rare (MAF < 1%) GWAS Tail Variants
POPout Effect Size
c
Age At Menopause
PRS
Lower Tail Regression
Trait Quantile
Effect Size
MAF
d
No Selection (9) (β = 0, γ = 0)
Positive (30) (β > 0)
Negative (27) (β < 0)
Stabilising (43) (γ < 0)
Phosphate
Haemoglobin Conc
RBC Dist Width
Sitting Height
Fluid Intelligence
HandGrip Strength
Match ID Time
IGF-1
Phosphate
Menopause
Pulse Rate
Heel BMD
HRT
BMR
Glucose
Body Fat %
Lifetime Reproductive Success
Residualized Trait
e
Model Selection: y = x_o + βX + γX^2
No Selection (β = 0, γ = 0)
Positive (β > 0)
Negative (β < 0)
Neutral Stabilising (β = 0, γ < 0)
Linear
Disruptive
Stabilising
Lower Tail Reg.
Upper Tail Reg.
L. Tail Reg.
U. Tail Reg.
Trait Effect
Not Sig (FDR 5%)
Mean Effect & 95% CI
Sig. Larger L. Tail Effect (Pv < 0.001)
Sig. Larger U. Tail Effect (Pv < 0.05)
f
Tail Rank Correlations: PopOut and Phenotype
Childred Fathered
Livebirths (Maternal)
Number of Illnesses
MisCarriages & StillBirths
Paternal Age
|*pval < 0.05, **pval < 0.01, ***pval < 0.001|